| Item | Standard | Limit |
|---|---|---|
| **Rear wheel** | | |
| Wheel type | Panel wheel *1 | ---- |
| | Cast wheel *2, *3, *4 | ---- |
| Rim size | 12 × 7.5 AT | ---- |
| Rim material | Steel *1 | ---- |
| | Aluminum *2, *3, *4 | ---- |
| Max. radial wheel runout | ---- | 2.0 mm (0.08 in) |
| Max. lateral wheel runout | ---- | 2.0 mm (0.08 in) |
| **Front disc brake** | | |
| Type | Dual | ---- |
| Disc outside diameter × thickness | 200.0 × 3.5 mm (7.87 × 0.14 in) | ---- |
| Brake disc minimum thickness | 3.0 mm (0.12 in) | ---- |
| Brake disc maximum deflection | 0.1 mm (0.004 in) | ---- |
| Pad thickness inner | 5.2 mm (0.20 in) | 1.5 mm (0.06 in) |
| Pad thickness outer | 5.2 mm (0.20 in) | 1.5 mm (0.06 in) |
| Master cylinder inside diameter | 17.46 mm (0.69 in) | ---- |
| Caliper cylinder inside diameter | 27.00 mm (1.06 in) | ---- |
| Recommended fluid | DOT 4 | ---- |
| **Rear disc brake** | | |
| Type | Dual | ---- |
| Disc outside diameter × thickness | 184.6 × 3.5 mm (7.27 × 0.14 in) | ---- |
| Brake disc minimum thickness | 3.0 mm (0.12 in) | ---- |
| Brake disc maximum deflection | 0.1 mm (0.004 in) | ---- |
| Pad thickness inner | 5.2 mm (0.20 in) | 1.5 mm (0.06 in) |
| Pad thickness outer | 5.2 mm (0.20 in) | 1.5 mm (0.06 in) |
| Caliper cylinder inside diameter | 25.40 mm (1.00 in) | ---- |
| Recommended fluid | DOT 4 | ---- |

| Item | Standard | Limit |
|---|---|---|
| **Parking brake** | | |
| Type | Disc | ---- |
| Disc outside diameter × thickness | 165.0 × 3.5 mm (6.50 × 0.14 in) | ---- |
| Brake disc minimum thickness | 3.0 mm (0.12 in) | ---- |
| Brake disc maximum deflection | 0.1 mm (0.004 in) | ---- |
| Brake pad lining thickness-inner | 3.2 mm (0.13 in) | 1.0 mm (0.04 in) |
| Brake pad lining thickness-outer | 3.2 mm (0.13 in) | 1.0 mm (0.04 in) |
| **Brake lever and brake pedal** | | |
| Accelerator pedal free play | 0 mm (0 in) | ---- |
| Brake pedal free play | 0 mm (0 in) | ---- |
| Parking brake cable free play | The maximum free play is equal to one click of the parking brake lever. | ---- |

EBS01004

# ELECTRICAL SPECIFICATIONS

| Item | Standard | Limit |
|---|---|---|
| **System voltage** | 12 V | ---- |
| **Ignition system** | | |
| Ignition timing (B.T.D.C.) | 12°/1,400 r/min | ---- |
| Advancer type | Digital | ---- |
| **Transistorized coil ignition** | | |
| Crankshaft position sensor resistance/color | 459 ~ 561 Ω at 20 °C (68 °F)/ gray/black | ---- |
| **ECU (engine control unit)** | | |
| Model/manufacturer | F8T83871/MITSUBISHI | ---- |
| **Ignition coil** | | |
| Model/manufacturer | 2JN/YAMAHA | ---- |
| Minimum ignition spark gap | 6.0 mm (0.24 in) | ---- |
| Primary coil resistance | 2.16 ~ 2.64 Ω at 20 °C (68 °F) | ---- |
| Secondary coil resistance | 8.64 ~ 12.96 kΩ at 20 °C (68 °F) | ---- |
| **Spark plug cap** | | |
| Material | Resin | ---- |
| Resistance | 10.0 kΩ | ---- |
| **AC magneto** | | |
| Model/manufacturer | F004T39372/MITSUBISHI | ---- |
| Standard output | 14.0 V 33.5 A at 5,000 r/min | ---- |
| Stator coil resistance/color | 0.099 ~ 0.121 Ω at 20 °C (68 °F)/ white-white | ---- |
| **Rectifier/regulator** | | |
| Type | Semiconductor-short-circuit | ---- |
| Model/manufacturer | FH012AA/SHINDENGEN | ---- |
| No load regulated voltage (DC) | 14.2 ~ 14.8 V | ---- |
| Rectifier capacity | 50.0 A | ---- |
| Withstand voltage | 40.0 V | ---- |
| **Electric starting system** | | |
| Type | Constant mesh | ---- |
| **Starter motor** | | |
| Model/manufacturer | SM-13/MITSUBA | ---- |
| Power output | 0.80 kW | ---- |
| Armature coil resistance | 0.0250 ~ 0.0350 Ω at 20 °C (68 °F) | ---- |
| Brush overall length | 12.5 mm (0.49 in) | 5.00 mm (0.20 in) |
| Spring force | 7.65 ~ 10.01 N (780 ~ 1,021 gf, 27.54 ~ 36.03 oz) | ---- |
| Commutator diameter | 28.0 mm (1.10 in) | 27.0 mm (1.06 in) |
| Mica undercut (depth) | 0.70 mm (0.03 in) | ---- |

| Item | Standard | Limit |
|---|---|---|
| **Starter relay** | | |
| Model/manufacturer | RC19-080A/MITSUBA | ---- |
| Amperage rating | 180.0 A | ---- |
| Coil winding resistance | 4.18 ~ 4.62 Ω at 20 °C (68 °F) | ---- |
| **Fuel sender** | | |
| Sender unit resistance (full) | 19.00 ~ 21.00 Ω | ---- |
| Sender unit resistance (empty) | 137.00 ~ 143.00 Ω | ---- |
| **Fuel injection system relay** | | |
| Model/manufacturer | ACM33211/MATSUSHITA | ---- |
| Coil resistance | 96.0 Ω | ---- |
| **Radiator fan motor relay** | | |
| Model/manufacturer | ACM33211/MATSUSHITA | ---- |
| Coil resistance | 96.0 Ω | ---- |
| **Load control relay** | | |
| Model/manufacturer | ACM33211/MATSUSHITA | ---- |
| Coil resistance | 96.0 Ω | ---- |
| **Four-wheel-drive motor relay 3** | | |
| Model/manufacturer | ACM33211/MATSUSHITA | ---- |
| Coil resistance | 96.0 Ω | ---- |
| **Headlight relay** | | |
| Model/manufacturer | G8HN-1C4T-DJ/OMRON | ---- |
| Coil resistance | 105.0 Ω | ---- |
| **Four-wheel-drive motor relay 1** | | |
| Model/manufacturer | G8HN-1C4T-DJ/OMRON | ---- |
| Coil resistance | 105.0 Ω | ---- |
| **Four-wheel-drive motor relay 2** | | |
| Model/manufacturer | G8HN-1C4T-DJ/OMRON | ---- |
| Coil resistance | 105.0 Ω | ---- |
| **Circuit breaker** | | |
| Circuit breaker type | Fuse | ---- |
| Fuses | | |
| Main fuse | 40.0 A | ---- |
| Headlight fuse | 15.0 A | ---- |
| Signaling system fuse | 10.0 A | ---- |
| Ignition fuse | 10.0 A | ---- |
| Auxiliary DC jack fuse | 10.0 A | ---- |
| Fuel injection system fuse | 10.0 A | ---- |
| Four-wheel-drive motor fuse | 10.0 A | ---- |
| Backup fuse | 10.0 A | ---- |
| Radiator fan motor fuse | 25.0 A | ---- |
| Spare fuse | 25.0 A | ---- |
| | 15.0 A | ---- |
| | 10.0 A | ---- |

EBS01005

# TIGHTENING TORQUES

## ENGINE TIGHTENING TORQUES

| Item | Part name | Thread size | Q'ty | Tightening torque | | | Remarks |
|---|---|---|---|---|---|---|---|
| | | | | Nm | m · kg | ft · lb | |
| Cylinder head (exhaust pipe) | Stud bolt | M8 | 4 | 15 | 1.5 | 11 | |
| Cylinder head | Bolt | M9 | 4 | 35 | 3.5 | 25 | M |
| Cylinder head | Bolt | M9 | 2 | 38 | 3.8 | 27 | E |
| Cylinder head | Bolt | M6 | 2 | 10 | 1.0 | 7.2 | |
| Spark plug | — | M10 | 1 | 13 | 1.3 | 9.4 | |
| Oil check bolt | Bolt | M8 | 1 | 10 | 1.0 | 7.2 | |
| Cylinder | Bolt | M10 | 4 | 50 | 5.0 | 36 | See TIP. E |
| AC magneto rotor | Nut | M16 | 1 | 60 | 6.0 | 43 | E |
| Balancer driven gear | Nut | M18 | 1 | 80 | 8.0 | 58 | Use a lock washer. E |
| Thermostat cover | Bolt | M6 | 2 | 10 | 1.0 | 7.2 | |
| Cylinder head air bleed bolt | Bolt | M6 | 1 | 10 | 1.0 | 7.2 | |
| Valve adjusting screw | Nut | M6 | 4 | 14 | 1.4 | 10 | |
| Decompression assembly | Bolt | M7 | 2 | 20 | 2.0 | 14 | |
| Timing chain guide (intake side) | Bolt | M6 | 2 | 10 | 1.0 | 7.2 | LT |
| Timing chain tensioner cap | Bolt | M16 | 1 | 20 | 2.0 | 14 | |
| Timing chain tensioner | Bolt | M6 | 2 | 10 | 1.0 | 7.2 | |
| Bearing retainer (camshaft) | Bolt | M6 | 2 | 10 | 1.0 | 7.2 | LT |
| Camshaft sprocket cover | Bolt | M6 | 2 | 10 | 1.0 | 7.2 | |
| Tappet cover | Bolt | M6 | 8 | 10 | 1.0 | 7.2 | |
| Camshaft sprocket | Bolt | M7 | 2 | 20 | 2.0 | 14 | |
| Crankcase | Bolt | M8 | 3 | 26 | 2.6 | 19 | |
| | Bolt | M6 | 4 | 10 | 1.0 | 7.2 | |
| | Bolt | M6 | 8 | 10 | 1.0 | 7.2 | |
| Engine oil drain bolt | Bolt | M14 | 1 | 30 | 3.0 | 22 | |
| Oil filter cartridge | — | M20 | 1 | 17 | 1.7 | 12 | |
| Oil filter cartridge union bolt | Union bolt | M20 | 1 | 68 | 6.8 | 49 | E |
| Oil pipe (dipstick) | Bolt | M6 | 1 | 10 | 1.0 | 7.2 | |
| Oil delivery pipe 1 | Union bolt | M14 | 2 | 35 | 3.5 | 25 | |
| Oil delivery pipe 1 | Union bolt | M10 | 1 | 20 | 2.0 | 14 | |
| Oil delivery pipe 1 | Bolt | M6 | 1 | 10 | 1.0 | 7.2 | |
| Oil delivery pipe 2 | Bolt | M6 | 2 | 10 | 1.0 | 7.2 | |
| Oil delivery pipe 3 | Bolt | M6 | 2 | 10 | 1.0 | 7.2 | |
| Oil pump | Bolt | M6 | 3 | 10 | 1.0 | 7.2 | |
| Oil pump driven gear | Nut | M10 | 1 | 22 | 2.2 | 16 | |
| Timing chain guide | Bolt | M6 | 2 | 10 | 1.0 | 7.2 | LT |
| V-belt cooling duct 1 | Bolt | M6 | 2 | 7 | 0.7 | 5.1 | |
| Air filter case | Bolt | M6 | 4 | 7 | 0.7 | 5.1 | |

| Item | Part name | Thread size | Q'ty | Tightening torque | | | Remarks |
|---|---|---|---|---|---|---|---|
| | | | | Nm | m · kg | ft · lb | |
| Air intake duct | Bolt | M6 | 2 | 7 | 0.7 | 5.1 | |
| Muffler | Bolt | M8 | 2 | 20 | 2.0 | 14 | |
| Muffler bracket | Bolt | M8 | 4 | 30 | 3.0 | 22 | |
| Exhaust pipe | Nut | M8 | 4 | 20 | 2.0 | 14 | |
| Tailpipe | Bolt | M6 | 3 | 10 | 1.0 | 7.2 | |
| Radiator inlet pipe holder | Bolt | M6 | 1 | 7 | 0.7 | 5.1 | |
| Radiator outlet pipe holder | Bolt | M6 | 1 | 7 | 0.7 | 5.1 | |
| Oil cooler inlet pipe 1/oil cooler outlet pipe 1 holder | Bolt | M6 | 1 | 7 | 0.7 | 5.1 | |
| Oil cooler inlet pipe 2/oil cooler outlet pipe 2 holder | Bolt | M6 | 2 | 7 | 0.7 | 5.1 | |
| Oil cooler inlet pipe 1/oil cooler outlet pipe 1 and oil cooler | Bolt | M17 | 2 | 21 | 2.1 | 15 | |
| Oil cooler | Nut | M6 | 4 | 7 | 0.7 | 5.1 | |
| Radiator | Bolt | M6 | 2 | 7 | 0.7 | 5.1 | |
| Coolant reservoir | Bolt | M6 | 2 | 7 | 0.7 | 5.1 | |
| Water pump housing | Bolt | M6 | 2 | 10 | 1.0 | 7.2 | |
| Coolant drain bolt | Bolt | M6 | 1 | 10 | 1.0 | 7.2 | |
| Water pump air bleed bolt | Bolt | M6 | 1 | 10 | 1.0 | 7.2 | |
| Water pump outlet pipe | Bolt | M6 | 1 | 10 | 1.0 | 7.2 | |
| Water jacket joint | Bolt | M6 | 2 | 10 | 1.0 | 7.2 | |
| Crankshaft end accessing screw | Bolt | M36 | 1 | 10 | 1.0 | 7.2 | |
| Timing mark accessing screw | Bolt | M14 | 1 | 6 | 0.6 | 4.3 | |
| Drive belt cover | Bolt | M6 | 12 | 10 | 1.0 | 7.2 | |
| Drive belt case | Bolt | M6 | 8 | 10 | 1.0 | 7.2 | |
| V-belt cooling duct 1 bracket | Bolt | M6 | 2 | 10 | 1.0 | 7.2 | |
| Bearing housing (primary sheave assembly) | Bolt | M6 | 4 | 10 | 1.0 | 7.2 | |
| Bearing retainer (bearing housing) | Bolt | M6 | 1 | 10 | 1.0 | 7.2 | |
| AC magneto cover | Bolt | M6 | 11 | 10 | 1.0 | 7.2 | |
| AC magneto/crankshaft position sensor lead holder | Bolt | M5 | 2 | 7 | 0.7 | 5.1 |  |
| Starter one-way clutch | Screw | M8 | 3 | 30 | 3.0 | 22 |  |
| Clutch housing assembly | Bolt | M6 | 9 | 10 | 1.0 | 7.2 | |
| Clutch carrier assembly | Nut | M22 | 1 | 190 | 19.0 | 140 | Stake.  |
| Middle drive pinion gear nut | Nut | M22 | 1 | 190 | 19.0 | 140 | Stake. |
| Middle driven shaft bearing housing | Bolt | M8 | 4 | 32 | 3.2 | 23 |  |
| Middle drive shaft bearing retainer | Screw | M8 | 4 | 29 | 2.9 | 21 | Stake.  |
| Middle driven shaft bearing retainer | Nut | M55 | 1 | 80 | 8.0 | 58 | Left-hand thread  |

| Item | Part name | Thread size | Q'ty | Tightening torque | | | Remarks |
|---|---|---|---|---|---|---|---|
| | | | | Nm | m · kg | ft · lb | |
| Middle driven pinion gear bearing retainer | Nut | M60 | 1 | 130 | 13.0 | 94 | Left-hand thread |
| Universal joint yoke nut (middle gear side) | Nut | M16 | 1 | 150 | 15.0 | 110 | |
| Middle driven pinion gear bearing housing | Bolt | M8 | 4 | 25 | 2.5 | 18 | |
| Drive shaft coupling gear nut (middle gear side) | Nut | M16 | 1 | 190 | 19.0 | 140 | |
| Primary sheave assembly | Nut | M16 | 1 | 140 | 14.0 | 100 | |
| Secondary sheave spring retainer | Nut | M36 | 1 | 90 | 9.0 | 65 | |
| Secondary sheave assembly | Nut | M16 | 1 | 100 | 10.0 | 72 | |
| Shift lever cover | Bolt | M6 | 4 | 10 | 1.0 | 7.2 | |
| Shift lever 2 assembly | Bolt | M6 | 1 | 14 | 1.4 | 10 | |
| Shift drum stopper | Bolt | M14 | 1 | 18 | 1.8 | 13 | |
| Crankcase plug bolt | Bolt | M14 | 1 | 18 | 1.8 | 13 | |
| Select lever unit | Bolt | M8 | 3 | 15 | 1.5 | 11 | |
| Shift arm | Bolt | M6 | 1 | 14 | 1.4 | 10 | |
| Shift rod locknut (select lever unit side) | Nut | M8 | 1 | 15 | 1.5 | 11 | Left-hand thread |
| Shift rod locknut (shift arm side) | Nut | M8 | 1 | 15 | 1.5 | 11 | |
| Stator coil assembly | Bolt | M6 | 3 | 7 | 0.7 | 5.1 | |
| Crankshaft position sensor | Bolt | M5 | 2 | 7 | 0.7 | 5.1 | |
| Coolant temperature sensor | — | M12 | 1 | 18 | 1.8 | 13 | |
| Gear position switch | Bolt | M6 | 2 | 7 | 0.7 | 5.1 | |
| Reverse switch | — | M10 | 1 | 17 | 1.7 | 12 | |
| Speed sensor | Bolt | M6 | 1 | 10 | 1.0 | 7.2 | |

**TIP:**
Temporarily tighten the cylinder bolts to 15 Nm (1.5 m · kg, 11 ft · lb) and then tighten them to 50 Nm (5.0 m · kg, 36 ft · lb).